

Problem 1: Cab Fare Estimator

Question

Create a function:

```
calculateFare(distance, waitingTime, surge)
```

Rules:

- Base Fare = ₹50
- ₹12 per km
- ₹2 per minute waiting charge
- If surge is `true`, increase the total fare by **50%**
- If the final fare is greater than ₹1000, give **5% discount**
- Print the final fare.

E-commerce Coupon Engine

Write a function:

```
applyCoupon(totalAmount, coupon)
```

Coupons

```
SAVE10  
10%
```

```
SAVE20  
20%
```

```
WELCOME  
Flat ₹150
```

```
FREESHIP  
₹80 off
```

```
INVALID  
No discount
```

Return

```
Original Price  
Discount  
Final Price
```

Instagram Username Generator

Create a function:

```
generateUsername(firstName, lastName, birthYear)
```

Requirements

Generate a username in the following format:

```
firstName_lastName_lastTwoDigitsOfBirthYear
```

Rules

- Take **First Name**, **Last Name**, and **Birth Year** as input.
- The username should contain:
 - First Name
 - _
 - Last Name
 - Last **2 digits** of the birth year.
- Your program should work for **any valid birth year**, not just specific years.

Examples

Input:

```
firstName = "Mohit"  
lastName = "Nara"  
birthYear = 1994
```

Output:

```
Mohit_Nara94
```

Input:

```
firstName = "Rahul"  
lastName = "Sharma"  
birthYear = 2005
```

Output:

```
Rahul_Sharma05
```

Parking Lot Billing System

Create a function:

`calculateParkingBill(entryHour, exitHour, vehicleType)`

Rules:

- Calculate total parking hours.
- If the vehicle exits **after midnight**, calculate the parking duration accordingly.
- Bike = ₹20/hour
- Car = ₹50/hour
- SUV = ₹80/hour
- Parking above 8 hours gets a ₹100 discount.
- Parking above 24 hours is invalid.
- Print:
 - Total Hours
 - Vehicle Type
 - Final Bill